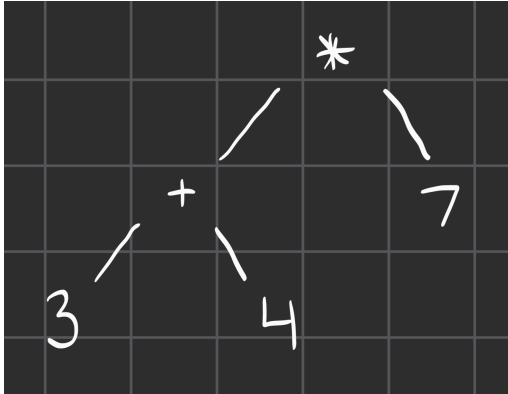


Programs as Data

- An expression tree is a structured way of modeling simple Python code



- By modelling programs as data, we can start to think about writing programs that operate on other programs
 - A Python interpreter is a program that runs Python code
 - A Java compiler is a program that turns Java code into a sequence of primitive instructions
 - PyCharm and PythonTA are programs that analyze Python code and report potential problems

Expressions to Statements

Expression: A unit of code that, when evaluated, produces a single value

Statement: A more general expression that can produce a value or have some other effect

- `x = 5`
 - `return 10`
 - `break`
 - `if` statements
 - loops
- Every expression is a statement, but not vice-versa

Variable Bindings

```
class Name(expr):  
    """ A variable name  
  
    === Attributes ===  
    id: The variable name
```

```
"""  
id: str
```

Variable Environment: A map from variable names to values. We will implement this using a Python `dict`

- `{'x': 1, 'y': True}`
- `Name('x').evaluate({'x': 10})`
 - The environment dictionary is passed as an argument to the `evaluate` method
- To create bindings, we use the `Assign` class
- Evaluating an `Assign` object mutates the environment

```
class Assign(Statement):  
    """ An assignment statement. <target> = <value>  
  
    === Attributes ===  
    target: the variable name  
    value: the expression  
    """
```

Summary

- `Name.evaluate` → Look up the variable name in the current environment
- `Assign.evaluate` → Add a new variable binding to the current environment